DIC25
Nonideal Effects in Dynamic Logic

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives

Complete the nine dynamic-logic experiments on handout PDF pages 99–106: charge sharing, explicit and parasitic coupling, leakage, three remedies, and power comparison at 500 MHz.

2 Handout requirements and test mapping

Requirement	Deck prefix and conditions	Status
Step 1: NAND charge sharing	01; A rises, B=0	Measured
Steps 2–4: CC sweep, inversion, move to output	02; 5–50 fF	Transition bracket observed
Steps 5–6: upper/lower NAND gate coupling	03; intrinsic coupling	Measured
Step 7: leakage during long evaluation	04; 20 us, 25/115 C	Measured
Step 8: always-on/feedback keeper	04/05/06; 500 MHz	Retention, sharing, power
Step 9: internal-node precharge	04/05	Sharing/leakage separated

3 Circuit and derivation

The dynamic NAND precharges at low clock and evaluates at high clock. An explicit 5 fF is placed at both output and internal stack nodes for reproducibility, in addition to MOS parasitics. A lossless sharing estimate is $V_f = (C_Y V_{Y0} + C_X V_{X0})/(C_Y + C_X)$, but threshold, body effect, initial levels, and feedthrough prevent exact agreement with an ideal equal-capacitor result.

A rapid aggressor transition approximately induces $\Delta V_Y \simeq C_C \Delta V_A/(C_Y + C_C)$. A driven inverter output has a restoring low-impedance path, whereas a floating dynamic node can retain erroneous charge. Keepers use weak PMOS devices with $W/L = 0.3/2\,\mu$m; the internal precharge PMOS uses $0.6/0.25\,\mu$m.

4 Simulation setup and interpretation

The supplied encrypted MOS model is evaluated at a baseline supply of 1.0 V and 25 °C. The 1.0 V value is a teaching condition, not a verified voltage rating. These results do not reproduce the handout's original process or supply conditions. Unless stated in the deck, $W_n/W_p/L = 0.6/1.8/0.25\,\mu$m, bulks are tied to their corresponding rails, and output load is 5 fF. Transmission-gate and weak-feedback dimensions are specified explicitly in each deck.

PrimeSim HSPICE S-2021.09 win64 was used with `RUNLVL=6`, `ACCURATE`, and `WARNLIMIT=100`. Every baseline and `.alter` analysis concluded without errors or failed measurements. The simulator still reports suppressed warnings associated with encrypted blocks; this does not establish that the model is warning-free or that those warnings are harmless. The `.tran` print interval is not a fixed integration step: the solver uses adaptive timesteps. A finite sweep is not a complete process qualification.

Unless stated otherwise, the declared functional diagnostic uses $V \leq 0.2$ V for low and $V \geq 0.8$ V for high. These are report-level interpretation thresholds, not numerical handout requirements or extracted receiver noise margins. Correct samples do not exclude glitches between samples. Figures are redrawn from actual simulation data; `results/` contains per-case waveforms, measurements, and derived comparison CSV files.

Public decks omit model loading paths; model configuration is added only to local execution copies. The model itself is not distributed. The first two filename digits identify the deck numbers used below. Every deck is standalone.

5 Results

In the basic sharing case without an output inverter, Y falls from 1.172805 to 0.701010 V, a 0.471795 V drop, while internal X reaches 0.622395 V. The initial value above 1 V is clock feedthrough, not a higher supply. Remedy comparisons use a separate set with matched inverter loading; their baseline droop differs and must not be mixed with this unloaded result.

Coupling (fF)	Dynamic Y (V)	Inverter output (V)	Direct-output minimum (V)
5	0.8277	0.0000	-0.2632
10	0.6555	0.0026	-0.4427
15	0.5511	0.0221	-0.5547
20	0.4845	0.2850	-0.6161
30	0.4027	0.9760	-0.6562
50	0.2859	0.9985	-0.6843

At CC=20 fF, the following inverter output is 0.284995 V, inside this report's conservative indeterminate region. At 30 fF it is 0.976020 V, a clear erroneous high. Using 0.5 V as a receiver decision level brackets the transition between 20 and 30 fF. The same 30 fF coupled directly to the inverter output briefly drives it to -0.656 V, but it recovers to approximately 0 V by 6 ns. Recovery within this window does not mean the spike is harmless to every receiver.

NAND gate position	Before (V)	After (V)	Drop (V)
upper	1.13912	0.94524	0.19388
lower	1.12564	1.07639	0.04925

With no added CC and only device parasitics, coupling at the upper NMOS gate produces a 0.193876 V Y drop, versus 0.049251 V at the lower gate. This supports the role of gate/source/drain

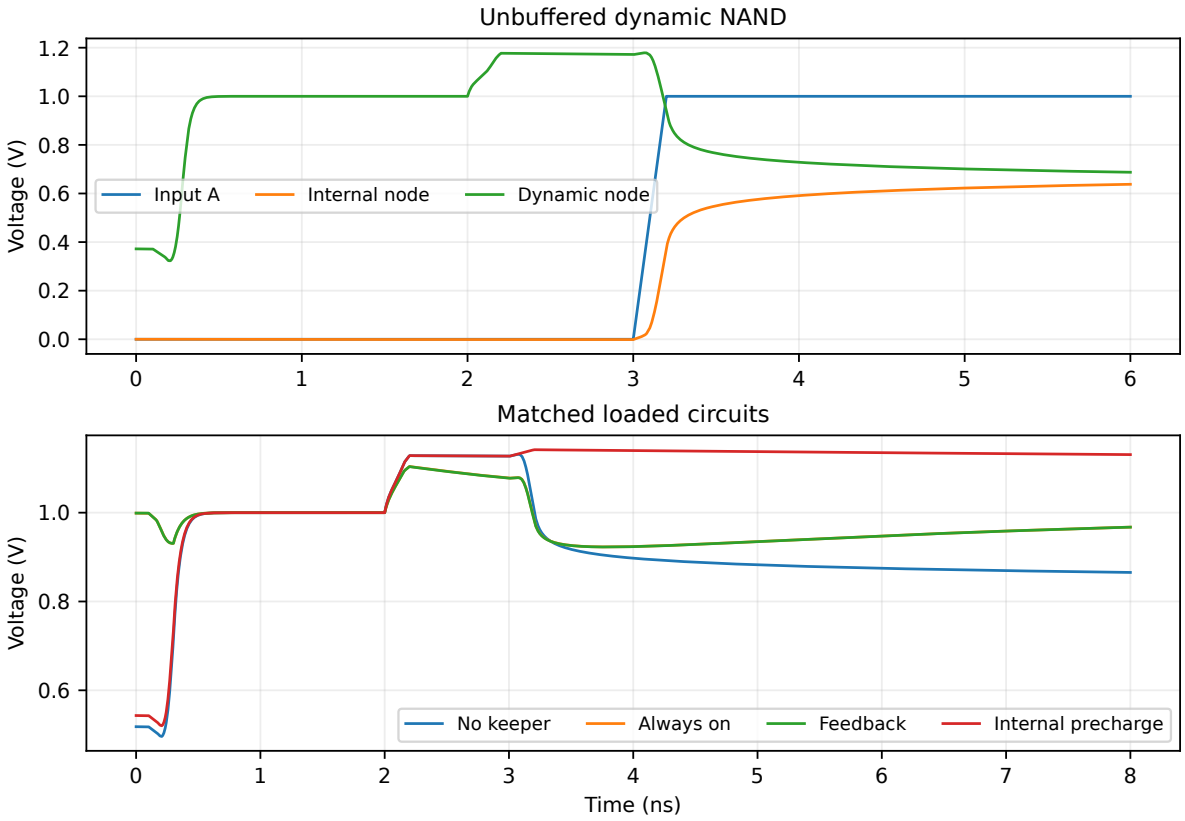


Figure 1: Basic sharing and the three remedies under matched loading.

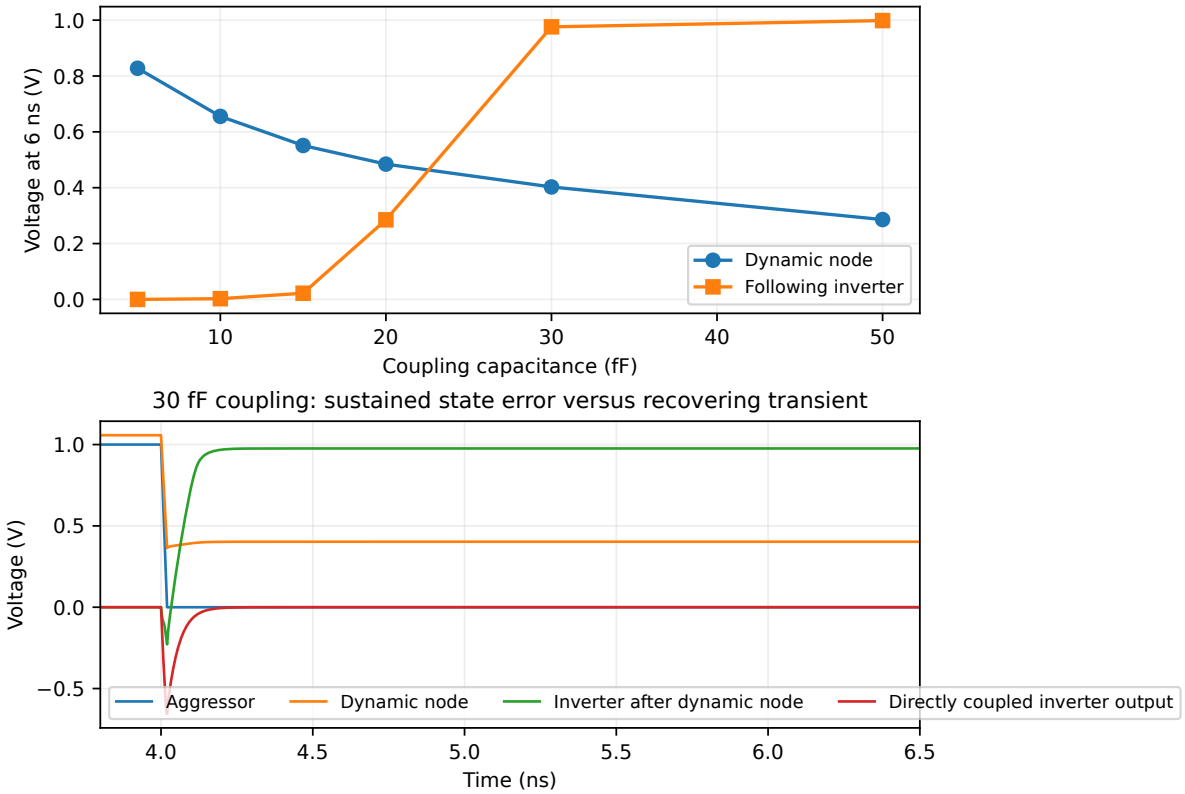


Figure 2: CC sweep and representative 30 fF waveforms, distinguishing sustained error from a recovering spike.

location and intermediate-node shielding: logically equivalent input pins need not have equal analog sensitivity.

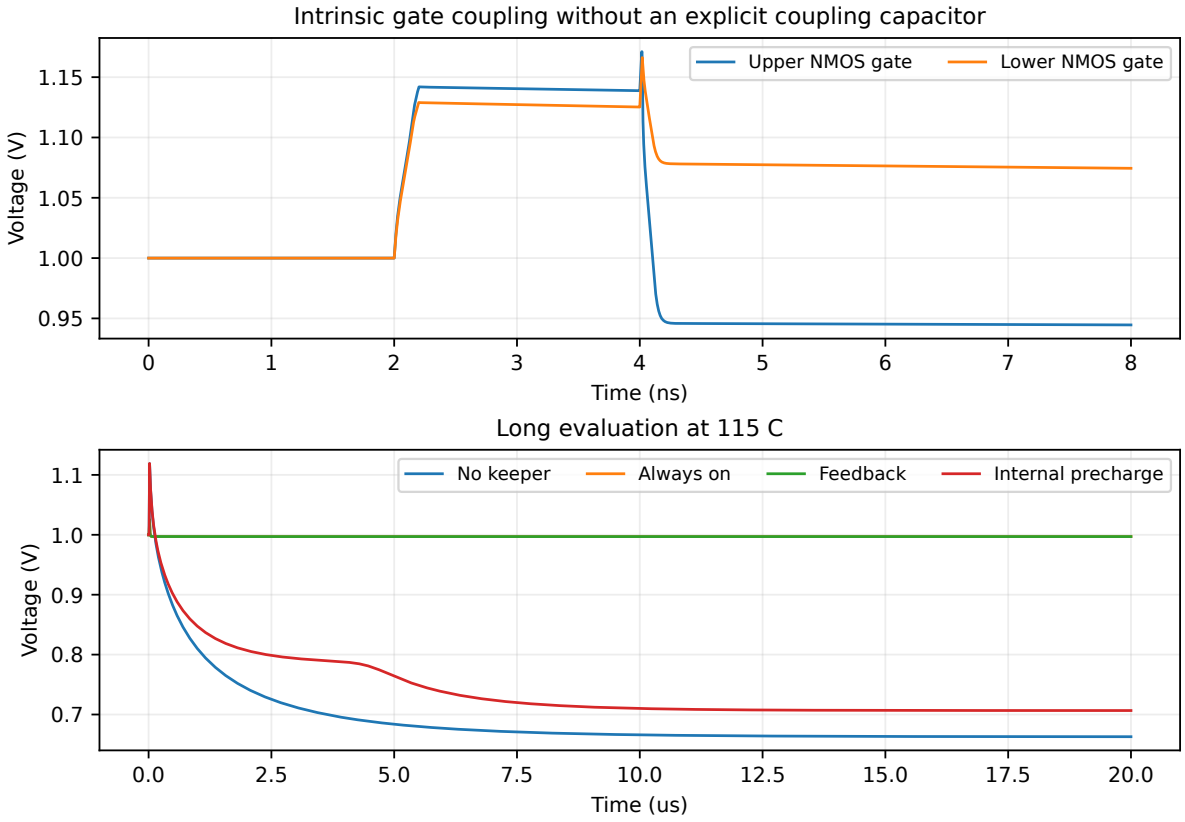


Figure 3: Upper: intrinsic parasitic coupling. Lower: 20 us retention at 115 C.

Remedy	Sharing drop (V)	Final at 25 C (V)	Final at 115 C (V)
None	0.24463	0.53783	0.66294
Always on	0.14851	0.99896	0.99702
Feedback	0.14830	0.99896	0.99702
Internal precharge	-0.01016	0.58577	0.70663

Always-on and feedback keepers maintain long-hold levels near 0.997–0.999 V. Internal precharge suppresses sharing but does not continuously replenish charge during evaluation. The uncorrected final level is higher at 115 C than at 25 C, showing that net current depends on multiple paths in this model and bias condition. It must not be rewritten as necessarily faster decay at higher temperature. A complete inverter logic flip was not observed within 20 us in this leakage test.

Freq. (MHz)	Always power (uW)	Feedback power (uW)	Always eval. (uA)	Feedback eval. (uA)
500	16.1263	16.5449	9.9347	8.2381
50	2.4854	1.6582	1.8482	0.6664

6 Discussion, differences, and unverified scope

At 500 MHz, feedback average power is 16.5449 μW, slightly above 16.1263 μW for always-on. Feedback must not be declared universally lower-power. Its evaluation current is lower, but cycle-average power also includes dynamic charging, delay, and switching contention. At 50 MHz, feedback

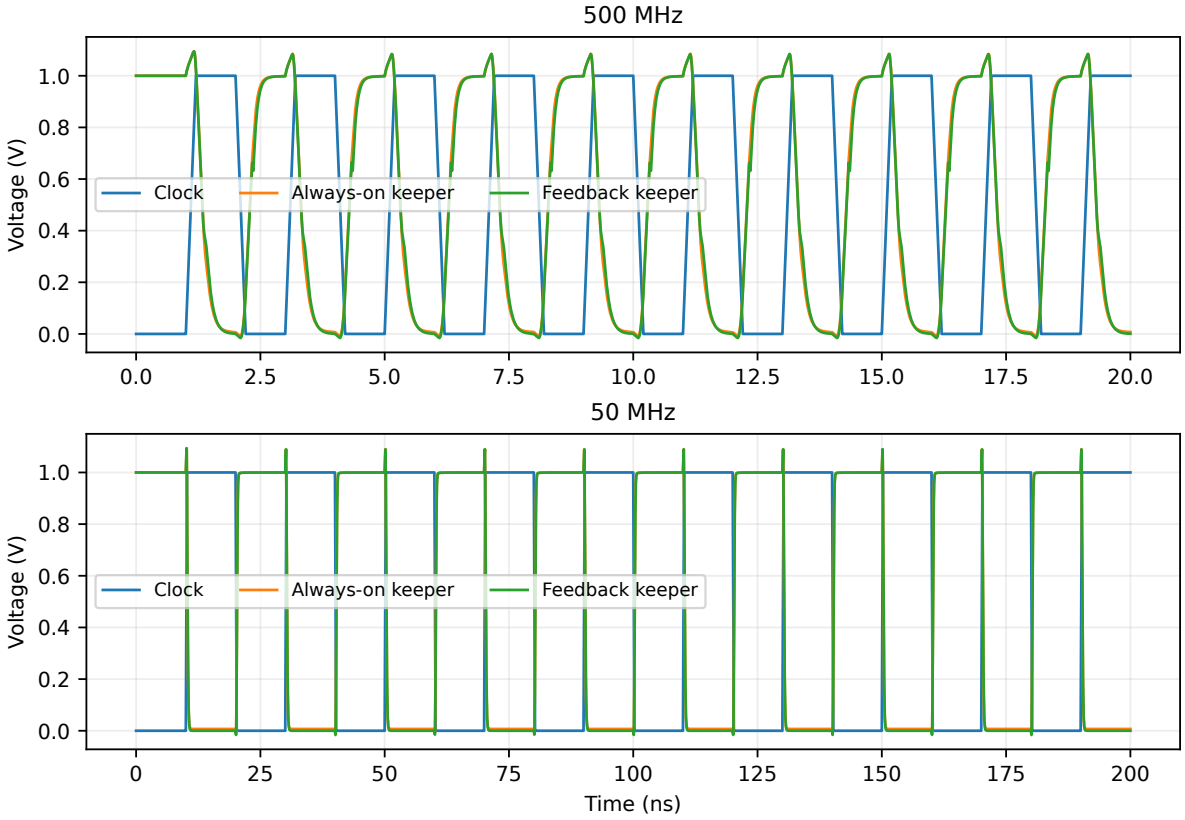


Figure 4: The required 500 MHz test is retained; 50 MHz is added as a more fully settled comparison.

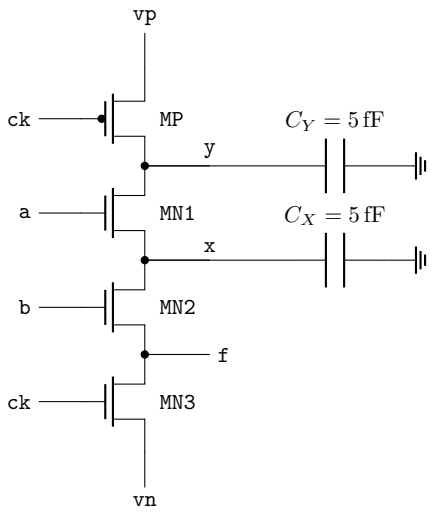
average power falls to 1.6582 μW versus 2.4854 μW for always-on. Power is averaged over $2T$–$10T$ and evaluation current over $8.7T$–$8.9T$. All nine observations have corresponding tests; qualitative handout explanations are not converted into invented numerical pass/fail requirements.

7 MOS transistor schematic

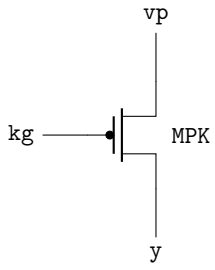
Source netlist: `01_charge_sharing.sp; 05_sharing_remedies.sp`. Dynamic NAND stack and the three charge-sharing remedies.

Identical node labels denote electrical connections. The drawings are editable CircuitikZ source; hierarchical boxes below refer to the MOS cells shown here.

Unless explicitly labeled otherwise, every NMOS body connects to `vn`=0 and every PMOS body to `vp`=V_{DD}; body wires are omitted for readability. Ordinary inverter and dynamic/clocked cells use $W_n/L_n = 0.6/0.25\,\mu$m, $W_p/L_p = 1.8/0.25\,\mu$m. Both TG devices use $W/L = 0.6/0.25\,\mu$m.



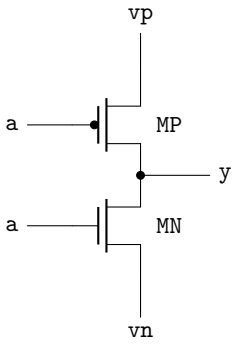
`dyn(a,b,ck,y,x,vp,vn)`



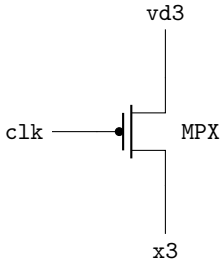
`MPK`: $W/L = 0.3/2\,\mu$m.

The base charge-sharing run has `b`=0 and $V_{DD} = 1$ V; its initial `x`=0 V can receive charge from precharged `y` when `a` rises. The capacitor values are explicit netlist elements. The keeper on the right belongs to the remedy variants; it is not present in the base run.

7.1 Keeper and internal-node precharge variants



`XI`: INV(y,out)



`MPX`: $W/L = 0.6/0.25\,\mu$m.

The inverter above uses input `a=y` and output `y=out` in the generic cell. Each `keep` instance contains one `dyn`, this inverter, and `MPK`. A behavioral voltage source `EG` selects the keeper gate voltage; it adds no MOS device.

Instance	Actual connections in the remedy deck
X0, MODE=0	$kg = vp$: keeper off. Nodes y0,x0,o0; supply vd0.
X1, MODE=1	$kg = 0$: always-on weak PMOS. Nodes y1,x1,o1; supply vd1.
X2, MODE=2	$kg = out = o2$: feedback keeper. Nodes y2,x2,o2; supply vd2.
X3, MODE=0	Keeper off; additional MPX precharges x3. MPX: D=x3, G=clk, S=B=vd3.

All four copies share `a`, `b`=0, and `ck=clk`; each has its own 1 V source. The feedback keeper senses the dynamic output through a real two-MOS inverter.